invent what we call a *unit vector* in the direction that we want to study. By a unit vector we mean one whose dot product with itself is equal to unity. Let us call this unit vector **i**; then $\mathbf{i} \cdot \mathbf{i} = 1$. Then, if we want the component of some vector in the direction of **i**, we see that the dot product $\mathbf{a} \cdot \mathbf{i}$ will be $a \cos \theta$, i.e., the component of **a** in the direction of **i**. This is a nice way to get the component; in fact, it permits us to get *all* the components and to write a rather amusing formula. Suppose that in a given system of coordinates, x, y, and z, we invent three vectors: **i**, a unit vector in the direction x; **j**, a unit vector in the direction y; and **k**, a unit vector in the direction z. Note first that $\mathbf{i} \cdot \mathbf{i} = 1$. What is $\mathbf{i} \cdot \mathbf{j}$? When two vectors are at right angles, their dot product is zero. Thus

$$\begin{aligned} &\mathbf{i} \cdot \mathbf{i} = 1 \\ &\mathbf{i} \cdot \mathbf{j} = 0 \quad \mathbf{j} \cdot \mathbf{j} = 1 \\ &\mathbf{i} \cdot \mathbf{k} = 0 \quad \mathbf{j} \cdot \mathbf{k} = 0 \quad \mathbf{k} \cdot \mathbf{k} = 1 \end{aligned} \tag{1.24}$$

Now with these definitions, any vector whatsoever can be written this way:

$$\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} = a_z\mathbf{k}. \tag{1.25}$$

By this means we can go from the components of a vector to the vector itself.

This discussion of vectors is by no means complete. However, rather than try to go more deeply into the subject now, we shall first learn to use in physical situations some of the ideas so far discussed. Then, when we have properly mastered this basic material, we shall find it easier to penetrate more deeply into the subject without getting too confused. We shall later find that it is useful to define another kind of product of two vectors, called the vector product, and written as $\mathbf{a} \times \mathbf{b}$. However, we shall undertake a discussion of such matters in a later chapter.